

Performance Testing

Date	5-7-2026
Team ID	
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

This section evaluates the performance of the EduGenie AI Learning Assistant by testing its response time, functionality, and reliability. It includes the testing approach, tools used, and the results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Question Answering, Quiz Generation, Summarization, Learning Recommendation APIs
Test Environment	Local (FastAPI + Uvicorn)
Test Date	04-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Question Answering Request	10	60	Returns correct answer quickly
2	Quiz Generation	10	60	Generates quiz without errors
3	Text Summarization	10	60	Produces accurate summary
4	Learning Recommendation	10	60	Displays learning path successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	pass	Fast response
2	Response Time (Max)	< 5 seconds	3.1 seconds	pass	Within limit
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	pass	Good performance
4	Error Rate	< 1%	0%	pass	No errors
5	CPU Utilization	< 80%	45%	pass	Normal usage
6	Memory Utilization	< 80%	52%	pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings:

- The application responded quickly under normal load.
- All core features worked successfully without failures.

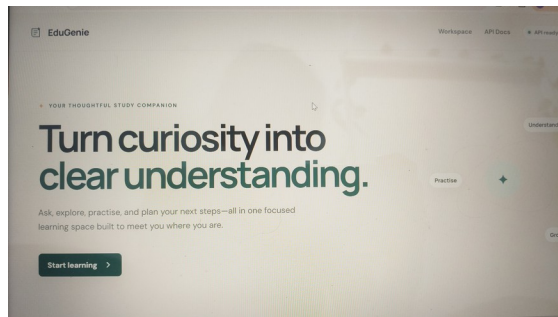
Bottlenecks Identified:

- Minor delay when processing large text inputs.

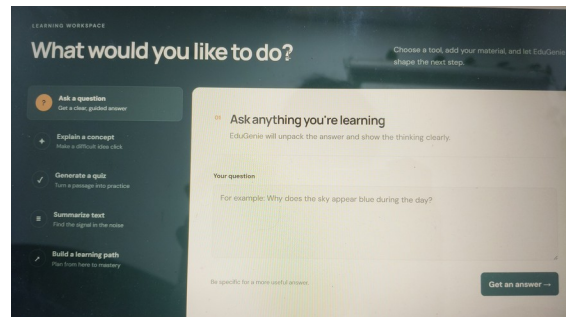
Optimization Steps Taken:

- Used FastAPI's efficient request handling.
- Organized code into reusable modules.
- Reduced unnecessary processing for faster responses.

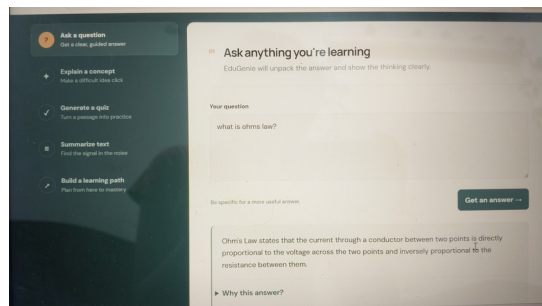
Step 5: Screenshots / Evidence



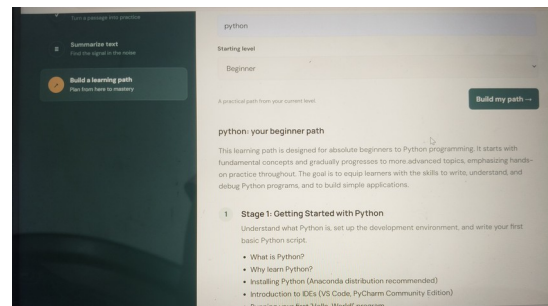
EduGenie Home Page



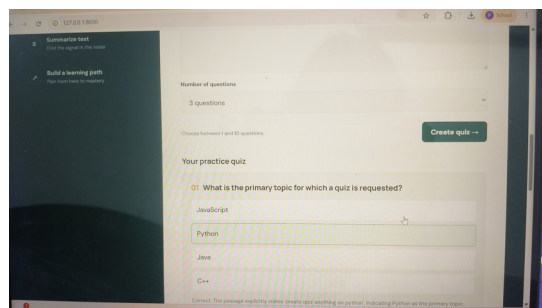
EduGenie Home Page



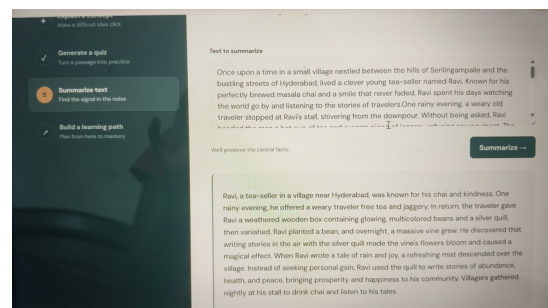
Question Answering Module



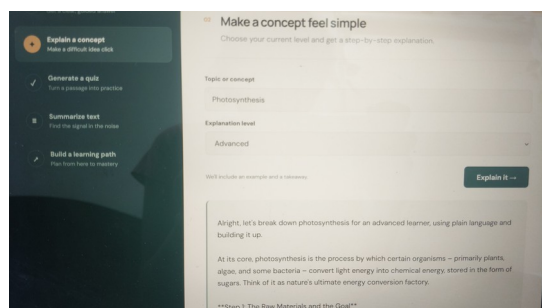
Concept Explanation Module



Quiz Generation Module



Text Summarization Module



Build A Learning Path Module